NLP

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Use Cases on Text Data

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Use cases on text data using Traditional NLP

There are multiple datasets which consist of free text data written by humans/machines which can be used for variety of use cases

- **Customer reviews** – Sentiment analysis, identify most occurring complaints
- **FAQ for users** – Build a chatbot which automatically suggests most relevant answer
- **Social media content** – Identify sentiment about the brand, identify top discussed topics
- **Server logs** – Identify /classify issues based on logs data

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Why Do We Need to Process Text for Machine Learning?

- **Machines don't understand text — they only understand numbers.** To teach a model to work with language, we must first **convert words into numbers** in a meaningful way.
- Raw text (like sentences or documents) must be transformed into numerical features.
- Unlike humans, ML models can't "read" — they process patterns in numbers.
- Techniques like TF-IDF, Word2Vec, etc., help us turn language into something models can learn from.
- This is the first step in Natural Language Processing (NLP).

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Text Data Cleaning

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Text data cleaning

- Convert the text to small case
- Remove punctuations and special characters
- Remove non-alphanumeric characters

➢ **Example:**

➢ **Input:** “ Hey Amazon – I have received a damaged mobile Applev4 at my pin code 12345. What can I do now? PLEASE REPLACE ASAP ! ”

➢ **Output:** hey amazon i have received a damaged mobile applev4 at my pin code what can I do now please replace asap

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Stop words removal

- Stop words are the words which are commonly use words in the sentence which do not provide any information e.g., a, an, the, that, this etc.
- These words should be removed from the data so that out model can focus on informative words
- Stop words list should be used carefully based on problem statement and enhanced based on domain in which you are working

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- **Example:**
  - **Input:** hey amazon i have received a damaged mobile applev4 at my pin code what can I do now please replace asap

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  - **Output:** hey amazon have received damaged mobile applev4 at pin code what can do now replace asap

Stemming

- Stemming is used to reduce words to their base form, also known as the root form.
- The process of stemming is used to normalize text and make it easier to process.
- A stemming algorithm reduces the words “retrieval”, “retrieved”, “retrieves” to the stem/root word “retrieve”

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- **Example:**
  - **Input:** hey amazon have received damaged mobile applev4 at pin code what can do now fix asap
  - **Output:** hey amazon have receive damage mobile applev4 at pin code what can do now fix asap

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Feature Extraction from Text Data

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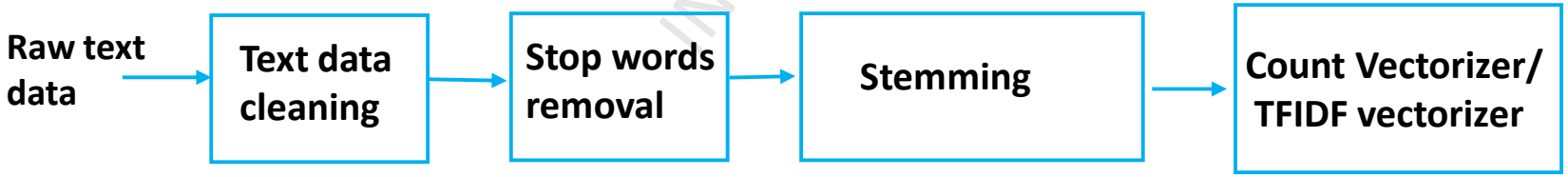
Features from text data

- Text data consists of words, but we cannot use it as input to machine learning models without converting it to numerical form
- There are multiple ways in which we can convert text data to numerical matrices. For e.g.
  - Bag of words (Count Vectorizer)
  - TFIDF Vectorizer



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Bag of words

- In bag of words (count vectorizer) feature value is represented by frequency of word in the document
- Features are decided by considering words in corpus (all documents)

Documents (cleaned text)	charger	heating	issue	network	cable	slow
charger heating issue	1	1	1	0	0	0
network cable issue	0	0	1	1	1	0
charger heating charger slow issue	2	1	1	0	0	1

Disadvantages of Bag of Words

- Every word is given the same weight/importance irrespective of its occurrence across documents

  Solution:

  ➢ Use TFIDF features, which give more importance to rare words

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TFIDF

In TFIDF features every word is given different weight depending on its occurrence in document and across corpus (all across documents)

- **TF:** Term Frequency (considers only current document)
- **IDF:** Inverse document frequency (considers whole corpus)

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Term Frequency

TF: Term Frequency

It is dependent on frequency of words in the given document

It can be calculated in various ways

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document d: charger heating charger slow issue

$f_{t,d}$: Frequency of term t in document d

n : Number of terms in document d (in above case its 5)

TF weight	TF value for word 'charger'
$f_{t,d}$	2
$f_{t,d}$/n	0.4
$\log_{10}(1+f_{t,d})$	0.48

Inverse Document Frequency

IDF: Inverse Document Frequency

It is dependent on frequency of words in the whole corpus

It can be calculated in various ways

Corpus (all documents)
charger heating issue
network cable issue
network cable broken
charger heating charger slow issue
slow speed issue

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document d: charger heating charger slow issue

n_t: Number of documents with term t

N : Number of documents in corpus (5 in this example)

IDF weight	IDF value for word 'charger'
$\log_{10}(N/n_t)$	0.40
$\log_{10}(N/(1+n_t))+1$	1.20
$\log_{10}((N-n_t)/n_t)$	0.18

TFIDF

- Suppose number of documents N = 5
- Then feature charger will have following value in TFIDF and Bag of Words (BoW)

TF : log (1+$f_{t,d}$)

IDF: $\log(N/(1+\ n_t))$ + 1

Documents (cleaned text)	charger TF	charger IDF	charger TFIDF	charger BoW
charger heating issue	0.3	1.20	0.36	1
charger heating charger slow issue	0.48	1.20	0.576	2

Note: Similarly, value for other features (words) can be calculated in TFIDF

Disadvantages of TFIDF

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Disadvantages of TFIDF and BoW

- For every new word added in corpus – feature size increases
- Word with similar meaning are considered as different words as semantic similarity is not considered

➢ Need a Solution which meets following criteria:

- Fixed vector size irrespective of number of words in corpus
- Semantic similarity between words is considered in feature representation

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Word2vec

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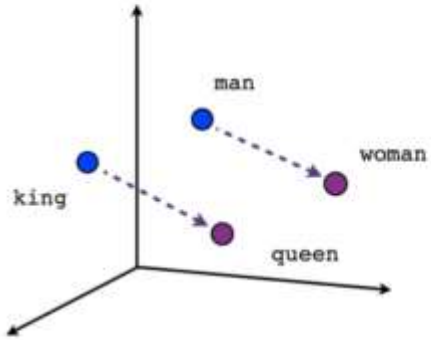
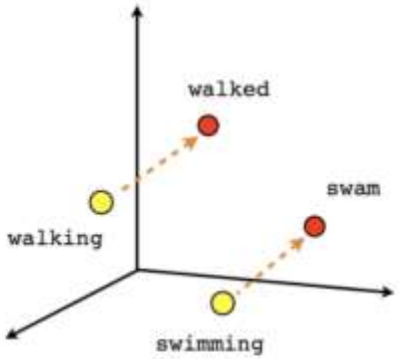
Word2vec

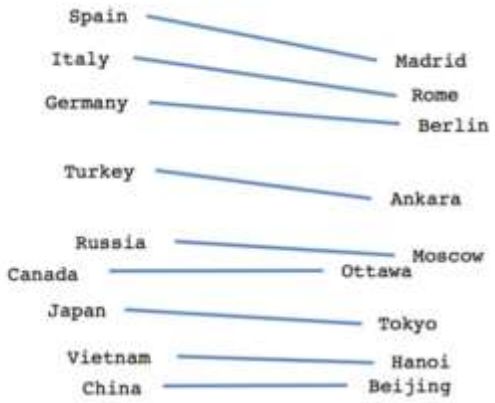
Word2vec takes large corpus of text as its input and produces a vector-space, with each unique word in the corpus being assigned a corresponding vector in the space.



Male-Female



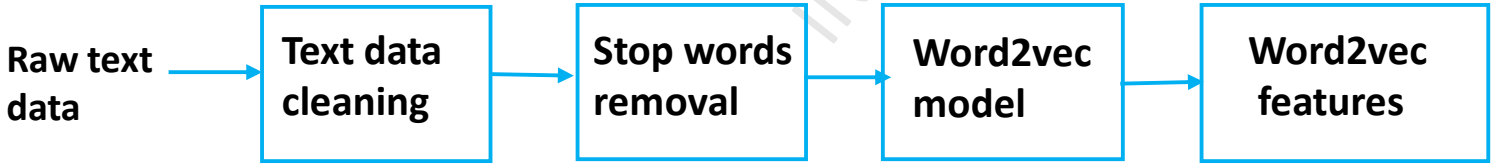
Verb tense



Country-Capital

Word2vec features from text data

- Text data consists of words, but we cannot use it as input to machine learning models without converting it to numerical form
- Word2vec models convert text data into numerical features of fixed dimension (typically 300) and features also consider semantic similarity between words



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Word2vec Architecture

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Word2Vec architectures

- **CBOW (Continuous Bag of Words) architecture**

- Predict center word by considering surrounding words as inputs

  received damaged cell phone delivery

- **Skip-gram architecture**

- Predict surrounding words by considering center word

  received damaged cell phone delivery

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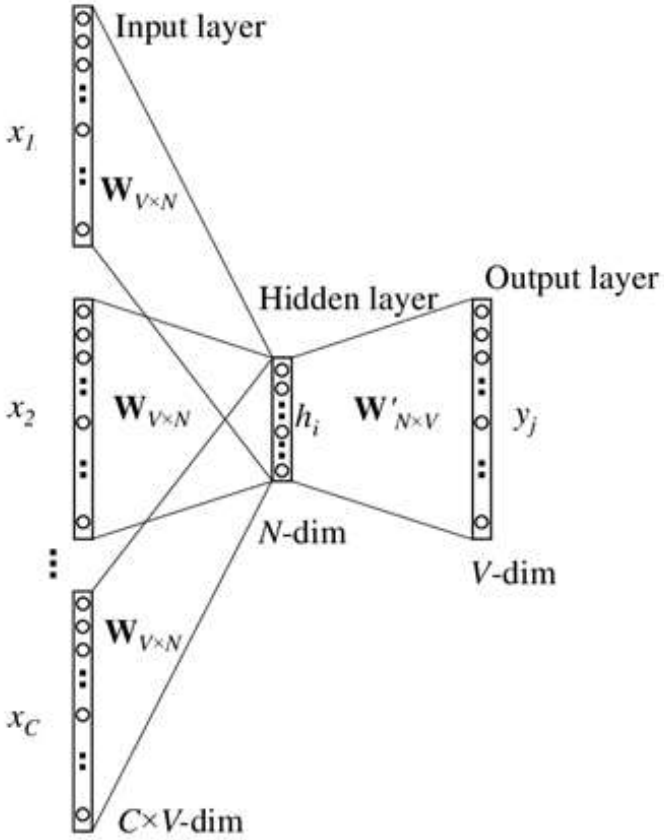
CBOW

Predict center word by considering surrounding words as inputs

C: context size
V: Vocabulary size (number of unique words)
N: word vector size



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What do we get after training Word2Vec model?

- Once we train a Word2Vec model on our text dataset, we get vector representations of words — called *word embeddings*
- Each word is represented by a vector of numbers (e.g., 100 or 300 dimensions).
- Words with similar meanings have similar vectors (they are close in space).

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```
"happy"     → [0.8, 0.3, 0.6]
"joyful"    → [0.79, 0.28, 0.61]
"sad"       → [-0.6, -0.2, 0.1]
```

Disadvantage of word2vec

Word2Vec cannot handle words it hasn’t seen during training.

What Does This Mean?

- If a word wasn't in the training data, Word2Vec **has no vector for it**.
- These unknown words are called **Out-of-Vocabulary (OOV)** words.
- The model **can’t represent them**, so you lose information when that word appears.

Why It's a Problem:

- You can't use those words as input features.
- This can hurt performance, especially in tasks like:
  - Sentiment analysis
  - Text classification
  - Chatbots

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FastText

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FastText

- FastText is an open-source library for text representation and classification developed by Facebook's AI Research team
- FastText breaks down the words into smaller subword units, such as character n-grams. By doing so, it can capture morphological information and handle out-of-vocabulary words efficiently

For e.g., walking can be broken down into subwords walk and ing. FastText learns embedding for subwords as well as original word

If the vocabulary has word eat but not the word eating – still FastText can get word embedding for eating by combining embedding for eat and ing

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Note: It also uses CBOW and Skip-gram architecture for training

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Questions?

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